

PVLoad-R1 Bench Card

Programmable resistive load for PV I–V characterisation · connections, power order, wiring

1 Arduino Uno → board 1×10 header

Pin	Arduino	Function
1	5V	Logic supply and relay coil supply (~30 mA total)
2	GND	Common reference
3	D6	K1 drive — bypasses R1 (470 kΩ)
4	D7	K2 drive — SHORT, bypasses the whole load
5	D8	K3 drive — bypasses DigiPot U2
6	D9	U2 chip select (CS#)
7	D10	U1 chip select (CS#)
8	D11	SPI MOSI → SDI on both chips
9	D12	SPI MISO ← SDO on both chips
10	D13	SPI SCK → both chips

Header pin 1 is the square pad on the silkscreen. Order down the header is 5V, GND, D6, D7, D8, D9, D10, D11, D12, D13. On the Uno, 5V and GND are on the power header; D8–D13 are in the digital group nearest the USB socket, D6 and D7 in the other group.

2 Banana jacks

Jack	Colour	Connect to
J1	Red	PV+ — cell positive (or the handheld's VΩ lead during bring-up)
J2	Red	+24 V from the bench supply
J3	Black	GND — supply return <i>and</i> cell negative. Shared.

J3 carries two or three things at once. Use a stacking banana plug; a jack with leads jammed alongside each other reads tens of ohms and drifts.

3 Instrument cables

Instrument	Cable, and what has to be set on the instrument itself
Arduino Uno	USB-A to USB-B (printer cable), straight to the laptop. Nothing to set.
Keithley 617 ×2	IEEE-488 (GPIB) only — no USB, no serial port. Each needs a USB-GPIB adapter , matched to the VISA installed on the laptop, which is National Instruments. One adapter serves both meters: IEEE-488 connectors stack, so run a cable from the adapter to the first meter and a second cable from the first meter to the second. Set a different primary address on each meter from its front panel before connecting them. They all ship at 27, so two on one bus will collide. Use 22 for the volts meter and 23 for the amps meter .
617 inputs	Front-panel triaxial INPUT, through a triax-to-dual-banana adapter. HI is the centre conductor, LO the inner shield.
Thorlabs EDFA100P	Plain USB cable — no adapter. It enumerates as a virtual COM port.

	The rear interlock must be shorted or the amplifier will not enable.
24 V supply	Two banana leads to J2 and J3. Set 24.0 V and a current limit of about 100 mA.

Every port and address above has a matching setting at the top of `PVLoad_Main.m`, described in the comments there.

Power sequence

Applies to every configuration. The cell and the 24 V supply are both sources.

THE ONE RULE

24 V goes on **FIRST** and off **LAST**. The cell goes on **LAST** and off **FIRST**.

The Arduino's 5 V pin is the board's 5 V rail — plugging in the Arduino is switching on 5 V.

Power up

#	Action	Notes
1	Seat both supply leads with the output off	Black to J3 first, then red to J2. No live plugging.
2	24 V supply output ON	Set 24.0 V, current limit ~100 mA. The board draws almost nothing here; the limit is a tripwire.
3	Arduino USB in — this is “5 V on”	USB alone is enough for ~30 mA of coils. The 9 V barrel jack is optional.
4	Power on both 617 electrometers	Any time before the script. Allow them to warm up.
5	Power on the EDFA100P	Interlock shorted. Output terminated or already aimed at the cell.
6	Connect the cell, dark	PV– to J3 first, then PV+ to J1. Nothing running; the board sits in OPEN by default, which keeps R1 in circuit.
7	Start the script	It enables the pump itself and illuminates the cell.

Power down

#	Action	Notes
1	Let the script finish, or Ctrl-C	The abort path kills the pump first, before anything else.
2	Confirm the pump is off	Check the EDFA front-panel ENABLE indicator before approaching the output.
3	Disconnect the cell	Darken it, then PV+ out of J1, then PV– out of J3.
4	Arduino USB out — 5 V off	
5	24 V supply output OFF	Then red out of J2, then black out of J3 if unplugging at all.
6	EDFA and 617s off	Any order.

WHY THE ORDER MATTERS

With 5 V up, K1 energised and 24 V absent, K1 shorts out R1 and the cell drives straight into the DigiPot's input clamp — about 80% of its ± 20 mA absolute maximum. That is the case that destroys a chip. D4 sits between the two rails and does *not* protect the PV input path.

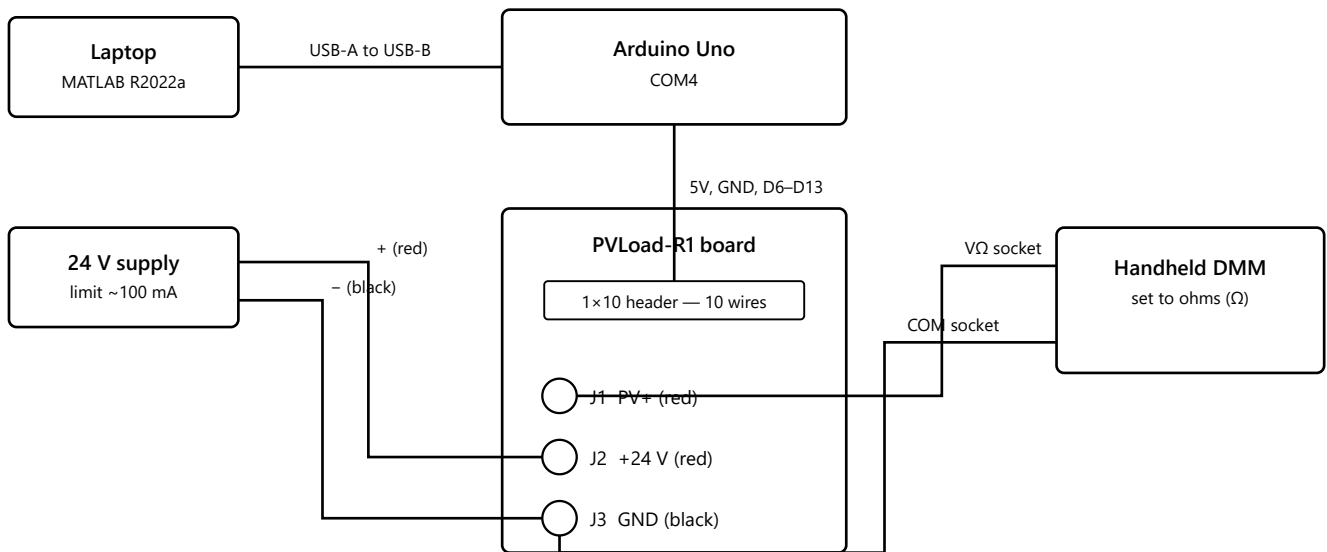
LASER

The EDFA100P is **CLASS 3B AT 1550 NM**, which is invisible. It emits up to 30 mW of amplified spontaneous emission with no optical input, at any pump current, whenever it is enabled. Terminate the output before running. Never look into the fibre bulkhead. If the interlock trips, it was opened by a person — software will not re-enable it and neither should you until you know why.

Shortcut for repeat runs: the jacks do not have to be cycled each time. A dark cell sources almost nothing, so blocking the light is equivalent to unplugging it and spares the connectors.

Wiring 1 · Handheld meter + 24 V

Board bring-up. No cell, no laser, no bench meters. The handheld stands in for the cell.



Both the supply return and the meter's COM lead land on J3. Use a stacking banana plug.

DO NOT put the red lead in the meter's 10A socket. In that socket the meter is a piece of wire and you would be shorting J1 to J3.

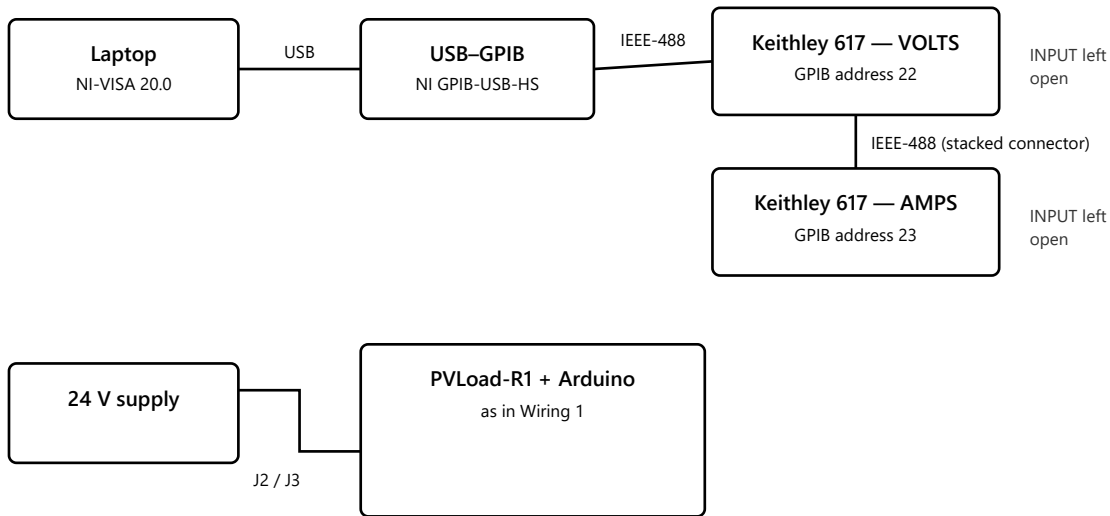
Power order for this setup

Up: 24 V output on → Arduino USB in → run. **Down:** stop → Arduino USB out → 24 V output off.

The meter leads stay plugged in throughout, both ways. In ohms mode a handheld is not a source and cannot hurt anything.

Wiring 2 • Two Keithley 617 + 24 V

Proving the GPIB chain and the 617 command dialect before a cell is involved. Meter inputs left open.



GPIB is a bus. One adapter serves both meters — IEEE-488 connectors stack, so the second cable runs meter to meter rather than back to the PC.

Before this works

- Each 617 must be on its own **primary address**, set from its front panel. They default to 27, which collides. Use 22 for volts and 23 for amps.
- The USB-GPIB adapter must be the same brand as the VISA runtime on the laptop. Mixing them is the usual reason nothing is found.
- Let the meters warm up before trusting a reading.

What it proves

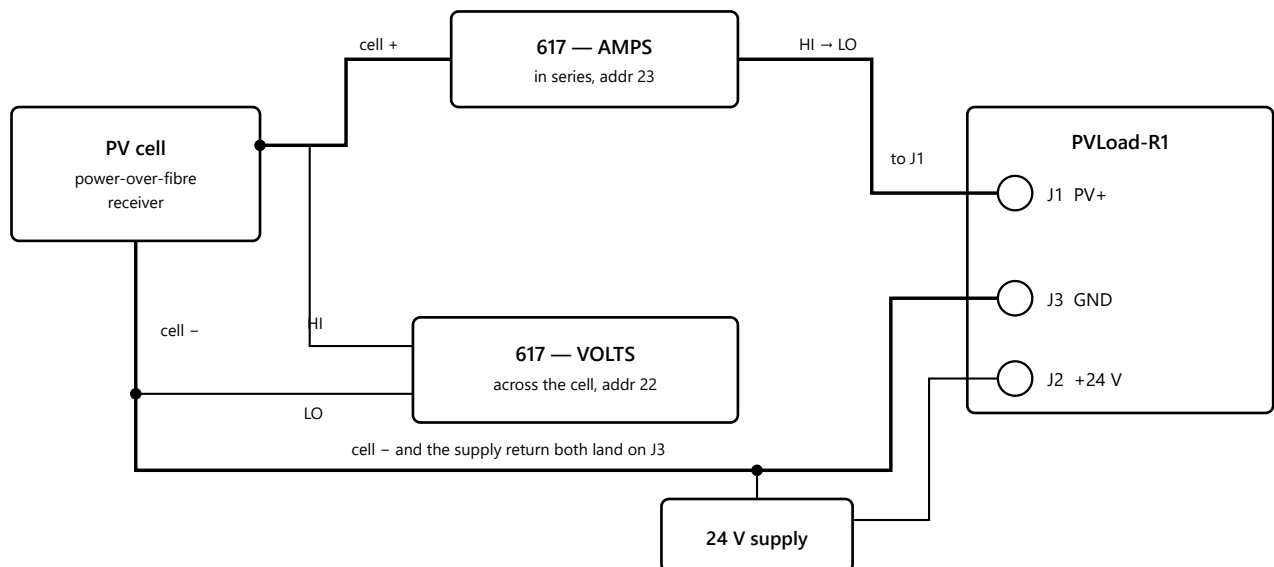
That the adapter, the cables and the two addresses are all good, and that the laptop can identify each meter, configure it and read from it. Everything except the measurement wiring, which Wiring 3 adds.

ABSOLUTE MAXIMUM ON THE 617 VOLTS INPUT IS 250 V PEAK. The amps function stops at 20 mA and there is nothing above it; a cell that outruns 20 mA needs an external shunt.

Wiring 3 • Full experiment

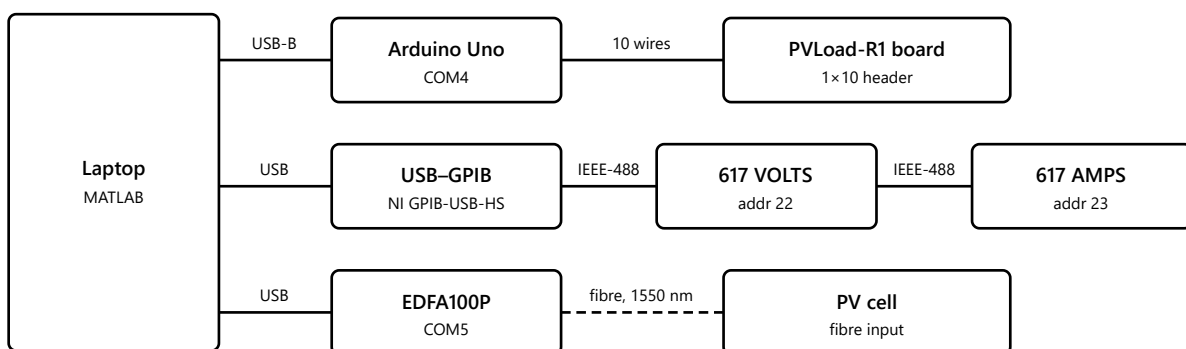
PV cell, both 617s, EDFA100P, 24 V supply, board. Everything connected at once.

3a The measurement loop



Thick lines carry the cell current. The ammeter breaks the PV+ lead outside the board; the voltmeter taps the cell terminals directly, **upstream of the ammeter**.

3b Cables back to the PC



Three independent chains. GPIB is a bus, so both 617s hang off one adapter with the second cable running meter to meter.

THE VOLTMETER GOES ON THE CELL SIDE OF THE AMMETER — directly on the cell leads, not across J1 and J3. Downstream of the ammeter it reads low by the burden drop. On a 617 that is under 1 mV on every range except 20 mA, where it is 3 mV, so it no longer swamps the Isc end — but the rule is unchanged.